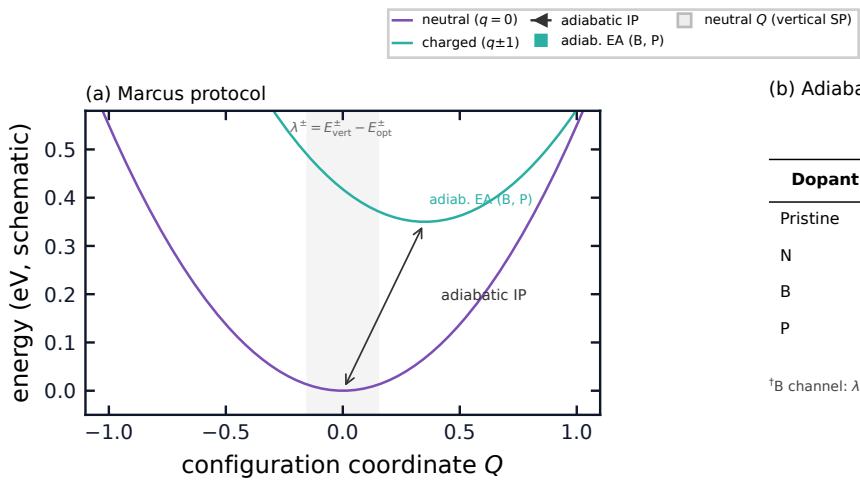

(b) Adiabatic IP/EA and $\lambda^{\pm}$

Dopant	IP (eV)	EA (eV)	λ^{+} (eV)	λ^{-} (eV)
Pristine	4.73	2.61	0.041	0.042
N	3.89	2.77	0.058	0.065
B	4.11	3.12	0.051	—[†]
P	4.07	2.96	0.200	0.315

[†]B channel: $\lambda^{-} < 0$ under fixed neutral-geometry vertical SP; not reported.